

Creative Automation Pipeline

A full-stack automation system for generating social media creative assets, featuring AI-powered content moderation, GenAI image generation, and a modern React frontend with AWS Cloudscape Design System.

Overview

This system automates the creation of social media creatives by:

- Providing a modern web interface for campaign management
- Accepting campaign briefs through forms or JSON API
- **AI-powered content compliance validation** using Groq's Llama 3.1 model
- **AI-powered hero image generation** using Replicate's Flux model when assets are unavailable
- Reusing existing assets when available
- Generating multiple aspect ratios (1:1, 9:16, 16:9) with img2img enhancement
- Adding campaign message overlays with responsive text
- Organizing outputs by product and aspect ratio
- Tracking campaign metrics and performance

review

1. **Configure API Keys** in `backend/app.py`:

Replace with your actual API keys

```
GROQ_API_KEY = "your_groq_api_key_here"
```

```
REPLICATE_API_TOKEN = "your_replicate_api_token_here"
```

2. **Run the complete stack:**

```
docker-compose up --build
```

The application will be available at

- Frontend: <http://localhost:3000>
- Backend API: <http://localhost:8000>
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Usage

Web Interface

1. **Access the frontend** at <http://localhost:3000>
2. **Upload Assets (Optional):**
 - Navigate to the Upload tab
 - Upload product images via the form interface
 - System automatically organizes assets by product name
3. **Create Campaign Brief:**
 - Fill out the campaign form with 2+ products
 - Add product names and descriptions
 - Set target region and audience
 - Enter your campaign message
4. **Submit and Monitor:**
 - Click "Generate Campaign"
 - **AI performs intelligent content compliance validation**
 - System reuses uploaded assets or **generates single hero image** if none available
 - **AI img2img model enhances assets** while maintaining aspect ratios
 - View real-time processing status and logs
 - Navigate to Downloads tab to access generated assets
5. **View Results:**
 - Browse generated creatives by campaign
 - Download individual assets or view metrics

API Usage

1. . Create Campaign Brief JSON

Create a JSON file with your campaign details

```
{
  "products": [
    {
      "name": "EcoClean Detergent",
      "description": "Environmentally friendly laundry detergent made from natural ingredients."
    },
    {
      "name": "FreshAir Fabric Softener",
      "description": "Natural fabric softener with essential oils."
    }
  ],
  "target_region": "North America",
  "target_audience": "Environmentally conscious families with children, ages 25-45",
  "campaign_message": "Clean your family's clothes the natural way. Gentle on skin, tough on stains, kind to the planet."
}
```

2. Submit Campaign

Using curl:

```
curl -X POST "http://localhost:8000/generate-campaign" \
-H "Content-Type: application/json" \
-d @example_campaign_brief.json
```

Response:

```
{
  "status": "accepted",
  "campaign_id": "a1b2c3d4",
  "message": "Campaign a1b2c3d4 is being processed"
}
{
  "status": "accepted",
  "campaign_id": "a1b2c3d4",
```

```
"message": "Campaign a1b2c3d4 is being processed"
}
```

3. Check Campaign Status

curl <http://localhost:8000/campaign/a1b2c3d4>

4. View Results

Generated creatives will be saved in:

output/

└── a1b2c3d4/

│ ├── ecoClean_detergent/

│ │ ├── ecoClean_detergent_1x1.jpg

│ │ ├── ecoClean_detergent_9x16.jpg

│ │ └── ecoClean_detergent_16x9.jpg

│ └── freshair_fabric_softener/

│ ├── freshair_fabric_softener_1x1.jpg

│ ├── freshair_fabric_softener_9x16.jpg

│ └── freshair_fabric_softener_16x9.jpg

Note: All variants maintain the hero image's original aspect ratio due to img2img model limitations.

Campaign Asset Generation Workflow

AI-Powered Asset Pipeline

The system uses a sophisticated AI pipeline to generate campaign assets:

1. **Hero Image Acquisition:**

- **Existing Assets:** Uses uploaded product images as hero images
- **AI Generation:** Creates single hero image using Flux Dev model if no assets available

2. **Aspect Ratio Variants via Img2Img:**

- **Input:** Single hero image (any aspect ratio)
- **Process:** AI img2img model enhances image quality while maintaining same aspect ratio
- **Models:** Uses bxc1ib2/flux_img2img for visual enhancement
- **Limitation: Maintains hero image's original aspect ratio** (does not change dimensions)

Text Overlay Application:

- **Python Rendering:** Responsive text overlays on each variant
- **Design:** Semi-transparent backgrounds for optimal readability
- **Content:** Product name and campaign message positioning

Benefits of Img2Img Approach

- **Visual Enhancement:** AI improves image quality and style consistency
- **Intelligent Processing:** Better than simple file copying
- **Efficiency:** Single hero → enhanced campaign assets
- **Quality:** Professional AI enhancement of source images

Img2Img Limitation

- ⚠ **Aspect Ratio Constraint:** The img2img model maintains the **same aspect ratio as the hero image**

- **Impact:** All generated variants (1:1, 9:16, 16:9) will have the hero image's original dimensions
- **Workaround:** Use hero images with desired aspect ratios, or combine with text-to-image generation

Asset Management

Method 1: Upload Assets via API

Upload product images through the web interface or API:

Via Web Interface:

- Navigate to the Upload tab in the frontend
- Select product name and image file
- Upload directly through the form

Via API:

```
curl -X POST "http://localhost:8000/assets/upload" \  
-F "product_name=EcoClean Detergent" \  
-F image=@path/to/your/image.jpg
```

Response:

```
{  
  
  "status": "success",  
  
  "message": "Image uploaded successfully for product 'EcoClean Detergent'",  
  
  "file_info": {  
  
    "filename": "image.jpg",  
  
    "size": 124567,  
  
    "path": "assets/ecoClean_detergent/image.jpg"
```

Method 2: Manual Asset Placement

assets/



Asset Discovery

The system automatically uses existing assets instead of AI generation. Check available assets:

curl <http://localhost:8000/assets/info>

API Endpoints

Endpoint	Method	Description
/	GET	Health check
/generate-campaign	POST	Submit campaign brief
/campaign/{campaign_id}	GET	Get campaign status/results
/campaigns	GET	List all available campaign IDs
/campaign/{campaign_id}/images	GET	List all generated images for a campaign

Endpoint	Method	Description
/campaign/{campaign_id}/download/{product_name}/{filename}	GET	Download a specific campaign image
/assets/upload	POST	Upload product assets via form (multipart/form-data)
/assets/info	GET	Get information about available assets
/metrics	GET	Get metrics for all campaigns

Key Design Decisions

1. Modular Architecture

Backend (FastAPI):

- `app.py`: FastAPI server orchestrating the pipeline
- `asset_manager.py`: Handles file organization and asset discovery
- `creative_generator.py`: AI-powered campaign asset generation using `img2img` model
- `image_generator.py`: AI-powered image generation
- `content_moderator.py`: AI-powered content validation (Groq Llama 3.1)
- `metrics_manager.py`: Campaign performance tracking

Frontend (React):

- `App.jsx`: Main application with AWS Cloudscape UI
- `CampaignBriefForm.jsx`: Campaign creation interface

- `CampaignResults.jsx`: Real-time status and results display
- `DownloadsPage.jsx`: Asset browsing and download interface
- `MetricsPage.jsx`: Campaign analytics and metrics

2. Asset Strategy

- **Primary**: Use existing assets from `/assets` folder as hero images
- **Fallback: Single AI-generated hero image using Replicate's Flux model**
- **Campaign Assets: AI img2img model enhances hero images** (maintains aspect ratios)
- **Text Overlays**: Python-based text rendering with responsive design

3. AI-Enhanced Asset Generation

- **Hero Image**: Single base image (existing asset or AI-generated)
- **Img2Img Enhancement**: AI model enhances image quality **maintaining original aspect ratio**
- **Limitation**: All variants have same aspect ratio as hero image
- **Quality**: AI improves visual appearance and consistency
- **Target Formats**: 1:1 (Instagram/Facebook), 9:16 (Stories/TikTok), 16:9 (YouTube/Covers)

4. Text Overlay Design

- Semi-transparent dark overlay at bottom for readability
- Responsive font sizing based on image dimensions
- Word wrapping to prevent text overflow
- Product name at top, campaign message at bottom

5. AI-Powered Compliance Pipeline

- **Pre-generation: AI-powered content validation using Groq's Llama 3.1 model**
- **Intelligent Analysis**: Context-aware policy violation detection for discriminatory content, illegal content, false claims, and excessive promotional language
- **Advanced Reasoning**: AI provides detailed explanations for policy violations instead of simple keyword matching

- **Failure Handling:** Campaigns fail safely with detailed AI reasoning and logging
- **Metrics Integration:** AI compliance results tracked and reported
- ☒ **Production Ready:** Uses sophisticated LLM for real-world content moderation

6. Asynchronous Processing

- Background tasks for campaign generation
- Non-blocking API responses
- Real-time status tracking with detailed logs
- Comprehensive error handling and recovery

AI Integration

Content Moderation (Groq)

- **Model:** Llama 3.1 8B Instant (free tier available)
- **Capabilities:**
 - Discriminatory content detection
 - Illegal content screening
 - False claims identification
 - Excessive promotional language analysis
- **Context-Aware:** Understands nuance vs. simple keyword matching
- **API:** Requires free Groq API key from console.groq.com

Hero Image Generation (Replicate Flux)

- **Model:** Flux Dev by Black Forest Labs
- **Quality:** Professional product photography with studio lighting
- **Customization:** Generates based on product name and description
- **Usage:** Only when existing assets unavailable (single image, not multiple)
- **API:** Requires Replicate API token from replicate.com

Campaign Asset Generation (Img2Img)

- **Model:** bxclib2/flux_img2img via Replicate
- **Process:** Enhances hero images while maintaining original aspect ratio
- **Quality:** AI-powered visual enhancement and style consistency
- **Parameters:** Configurable denoising (0.25), steps (20), sampling methods
- ⚠ **Limitation: Does not change aspect ratios** - maintains hero image dimensions
- **Efficiency:** Single hero image → enhanced campaign assets
- **API:** Uses same Replicate API token

Assumptions and Limitations

Assumptions

- **Minimum 2 products** required per campaign
- **English text** support (could be extended for localization)
- **Standard social media dimensions** for aspect ratios
- **JPEG output format** for broad compatibility
- **Ephemeral Usage:** Data persistence not required between container restarts
- **Development Environment:** Not intended for production use without significant enhancements

Current Limitations

1. **Storage & Data Persistence:**
 - **Ephemeral Storage:** All data (assets, outputs, metrics) disappears when containers are recreated
 - **No Volume Persistence:** Docker volumes not configured for data retention
 - **Local File System Only:** No cloud storage or database integration
2. **Compliance System (Mock Implementation):**
 - **No Brand Intelligence:** Brand compliance checks are in place
3. **Image Generation & Campaign Assets:**

- **Hero Images:** Single AI-generated image using Replicate's Flux Dev model (when assets unavailable)
- **Campaign Assets:** AI img2img model enhances images while maintaining hero's aspect ratio
- **⚠ Aspect Ratio Limitation:** Img2img model cannot change aspect ratios - all variants match hero dimensions
- **Professional Quality:** Both hero and enhanced variants produce studio-quality results
- **Generation Time:** Hero image 10-30 seconds, each img2img enhancement 10-120 seconds
- **Efficiency:** Single hero → enhanced campaign assets using AI processing
- Asset reuse prioritized over generation for cost and speed optimization

4. **Text Rendering:**

- Basic font support (system fonts only)
- Limited typography customization
- No brand font loading

5. **Processing & Scalability:**

- Single-threaded processing
- No batch processing optimization
- Limited error recovery
- No horizontal scaling capabilities

Future Enhancements

Critical Production Requirements

- **GenAI Hero Image Generation:** ☒ **COMPLETED** - Uses Replicate's Flux Dev model for professional image generation
- **AI Campaign Asset Generation:** ☒ **COMPLETED** - Uses img2img model for visual enhancement (maintains aspect ratios)
- **Data Persistence:** Docker volumes or database integration for data retention
- **AI-Powered Compliance:** ☒ **COMPLETED** - Uses Groq's Llama 3.1 for intelligent content validation

- **Computer Vision Compliance:** Image analysis for brand guideline validation (basic brand checks removed)
- **Cloud Storage:** S3/GCS integration with persistent asset management
- **Database Integration:** PostgreSQL/MongoDB for campaign and metrics storage

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Troubleshooting

Common Issues

Data Loss After Container Restart:

- **Issue:** All generated campaigns, assets, and metrics disappear when containers are recreated
- **Cause:** No persistent Docker volumes configured
- **Workaround:** Copy important data out of containers before stopping: `docker cp container_name:/app/backend/output ./backup`
- **Container Issues:** Ensure Docker and Docker Compose are installed and running
- **Font Issues:** System falls back to default fonts if custom fonts unavailable
- **Port Conflicts:** Check that ports 3000 and 8000 are available
- **AI API Dependencies:**

- **Issue:** Content moderation or image generation may fail
 - **Cause:** Missing or invalid API keys for Groq/Replicate
 - **Solution:** Ensure valid API keys are configured in `backend/app.py`
- **Rate Limiting:**
 - **Issue:** AI services may temporarily reject requests
 - **Cause:** Free tier rate limits on Groq or Replicate
 - **Solution:** Wait and retry, or upgrade to paid tiers
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Development

```

|—— docker-compose.yml      # Container orchestration

|—— example_campaign_brief.json # Sample campaign data

|—— test_pipeline.py         # Backend testing script

|—— README.md               # This documentation

|

|—— backend/                 # FastAPI backend

```

```
|   |—— app.py          # Main FastAPI application
|   |—— Dockerfile      # Backend container config
|   |—— requirements.txt # Python dependencies
|   |—— utils/          # Modular utilities
|   |   |—— asset_manager.py # Asset discovery and organization
|   |   |—— creative_generator.py # AI img2img campaign asset generation
|   |   |—— image_generator.py # AI image generation
|   |   |—— content_moderator.py # Compliance validation
|   |   |—— metrics_manager.py # Performance tracking
|   |—— assets/         # Input assets (optional)
|   |   |—— water/
|   |   |—— coca_cola/
|   |—— output/         # Generated creatives
|   |   |—— campaign_id1/
|   |   |—— campaign_id2/
|   |—— metrics/        # Campaign performance data
|
|—— frontend/          # React frontend
    |—— src/
        |—— App.jsx      # Main application
        |—— components/  # UI components
```

| └── services/ # API integration

|── Dockerfile # Frontend container config

|── package.json # Node.js dependencies

|── vite.config.js # Build configuration

└── nginx.conf # Production web server config